

Mapping Access to Olympic Investment Across Los Angeles

LA28 Equity Watch — Research Brief (draft)

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Draft findings language is generated from pipeline outputs for structure. Rewrite before Fulbright citation.

1. Motivation

Research question: How are LA28-related infrastructure investments and accessibility gains distributed across communities with different levels of socioeconomic disadvantage?

Los Angeles will host the 2028 Olympic and Paralympic Games amid Metro's Twenty-Eight by '28 program and a fixed venue map. Olympic legacy debates often claim equity benefits without measuring **how accessibility gains are distributed** across disadvantaged communities. This prototype builds a reproducible, tract-level equity analysis for Los Angeles County as the first stage of a research agenda culminating in Brisbane 2032 (with Hickman / UQ transport-accessibility methods).

Unit of analysis: 2010 census tracts, Los Angeles County (n=2346; complete-case n=2319).

2. Data & Methods

Full documentation: [METHODOLOGY.md](#). Checklist: [docs/FULBRIGHT_PREDECESSOR_CHECKLIST.md](#).

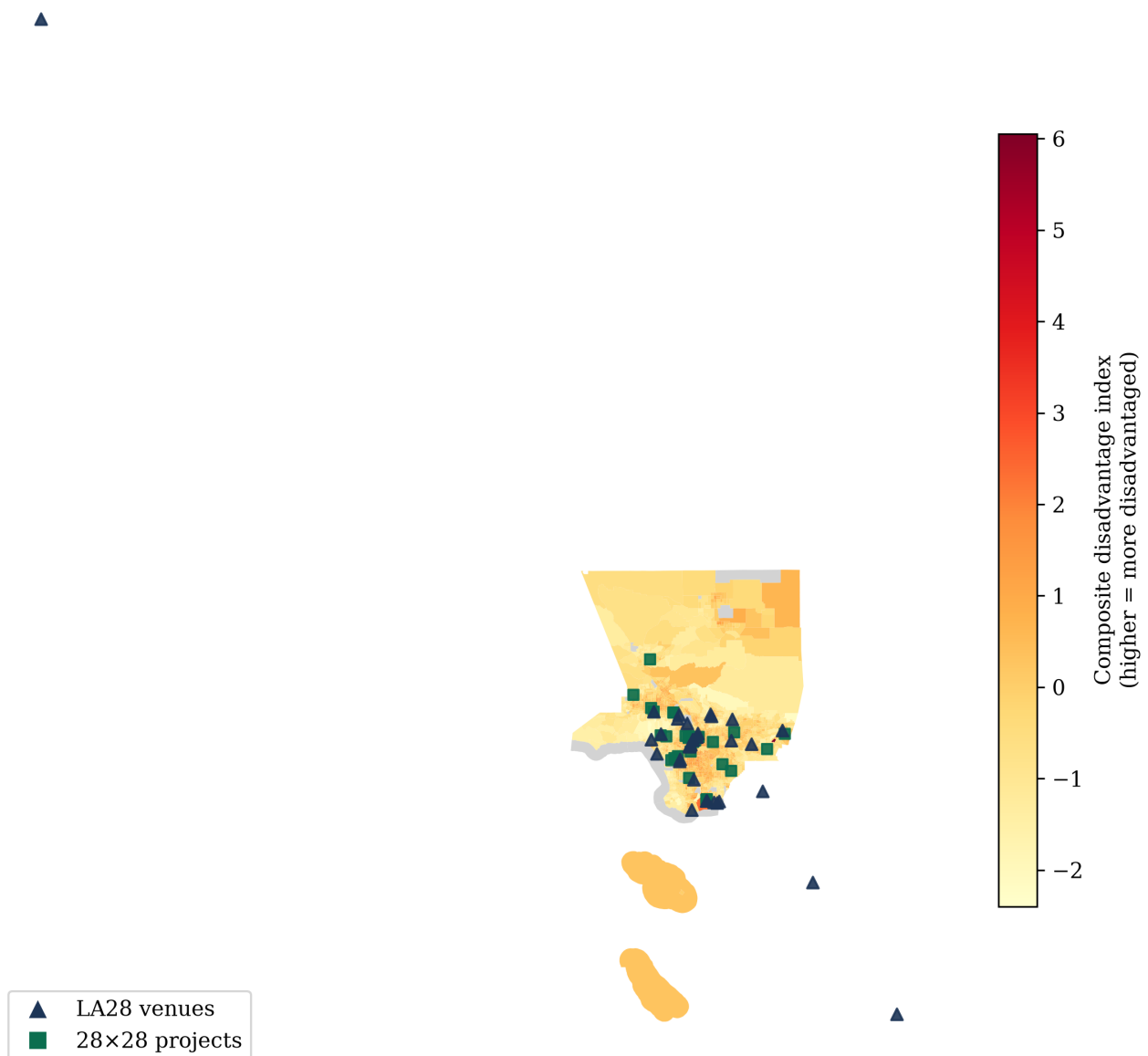
- **Disadvantage:** Equal-weight z-score composite of poverty, inverted income, no-vehicle share, rent burden $\geq 30\%$, overcrowding, and CalEnviroScreen percentile (race/ethnicity as covariates; HPI validation).
- **Investments:** Geocoded LA28 venues (competition + village) and Metro 28x28 points (transit, hubs, active transport, highway), each with documented sources.
- **Baseline access:** Network-adjusted walking minutes (detour factor 1.35) and GTFS **rail network** multimodal minutes.
- **Prospective scenarios:** S0 rail baseline; S1 opened+under construction; S2 full announced; S3 operational-only (delay).

- **Sensitivity:** Detour factors, 15/30/45-minute thresholds, facility definitions, CES vs composite.
 - **Benefits vs burdens:** Access benefits modeled; displacement/construction/security burdens named in [docs/BENEFITS_BURDENS_AND_COMMUNITY.md](#).
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3. Findings

Disadvantage and investment geography

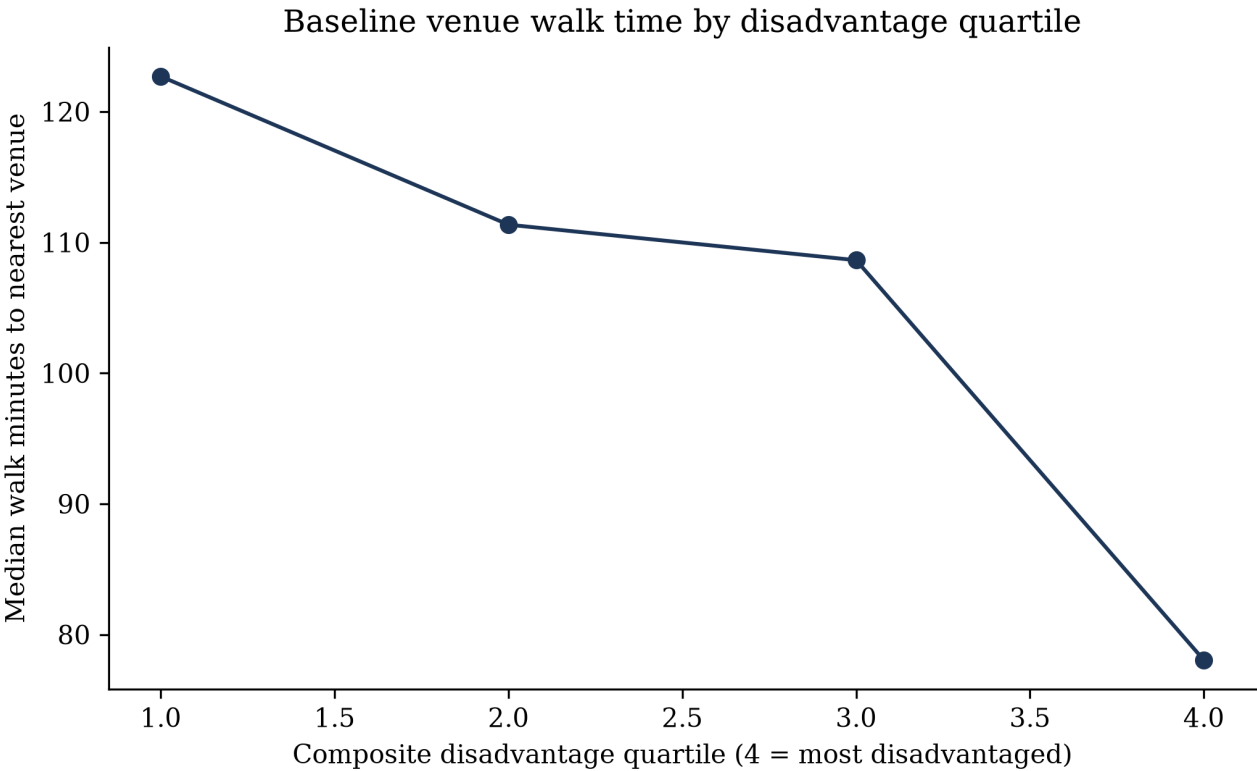
Composite socioeconomic disadvantage with LA28 venues
and Metro Twenty-Eight by '28 points



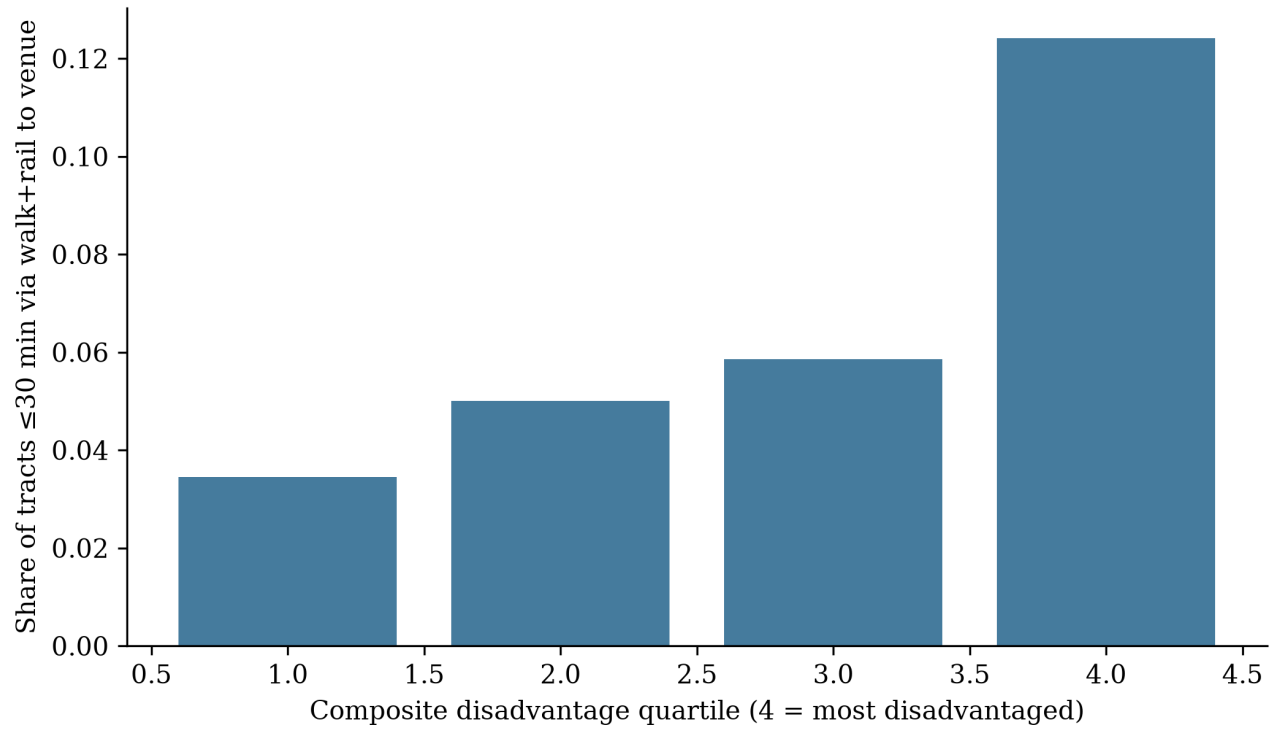
Baseline accessibility

Median network-adjusted walk time to the nearest LA28 venue is **122.7 minutes** in the least-disadvantaged quartile (Q1) versus **78.0 minutes** in the most-disadvantaged quartile (Q4) (Q4–Q1 = -44.6 min). Spearman ρ (disadvantage decile, walk minutes) = **-0.238** ($p = 3.43\text{e-}31$).

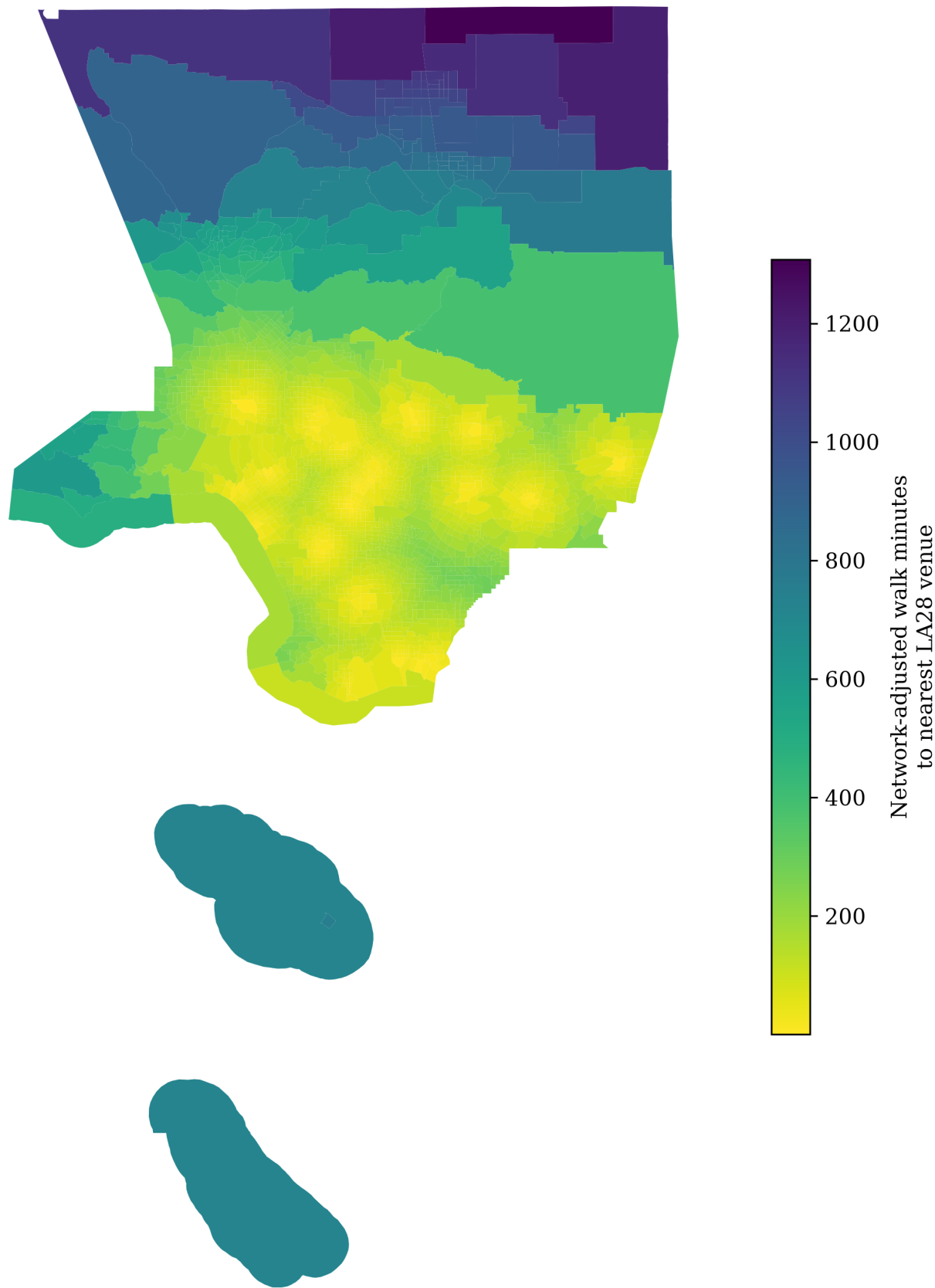
Share of tracts with ≤ 30 -minute **walk+rail** access to a venue-serving stop: **3.5%** (Q1) vs **12.4%** (Q4). Spearman ρ with disadvantage = **0.141** ($p = 7.70\text{e-}12$).



30-minute multimodal transit access to venues by quartile



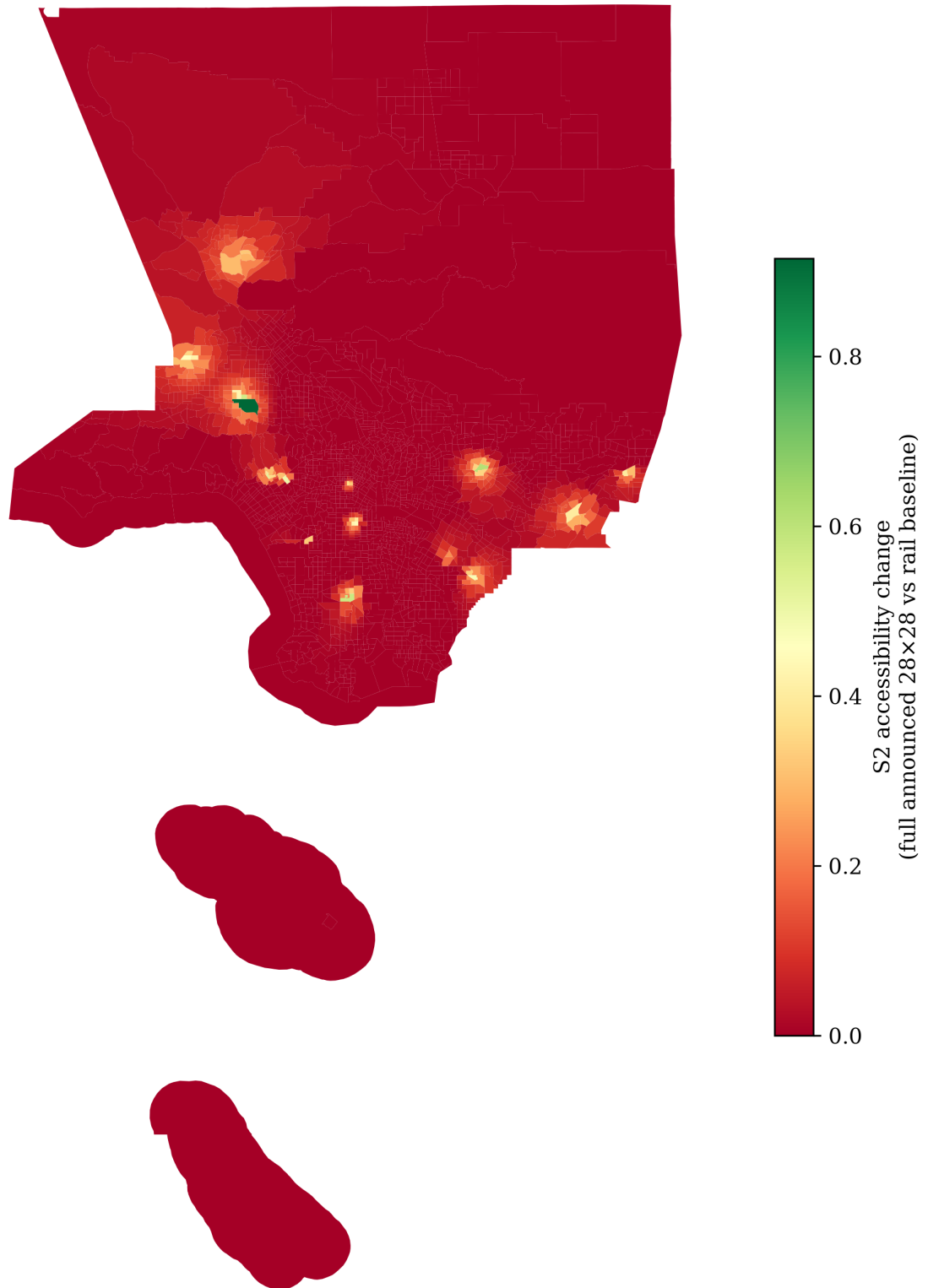
Baseline walking accessibility to nearest LA28 venue



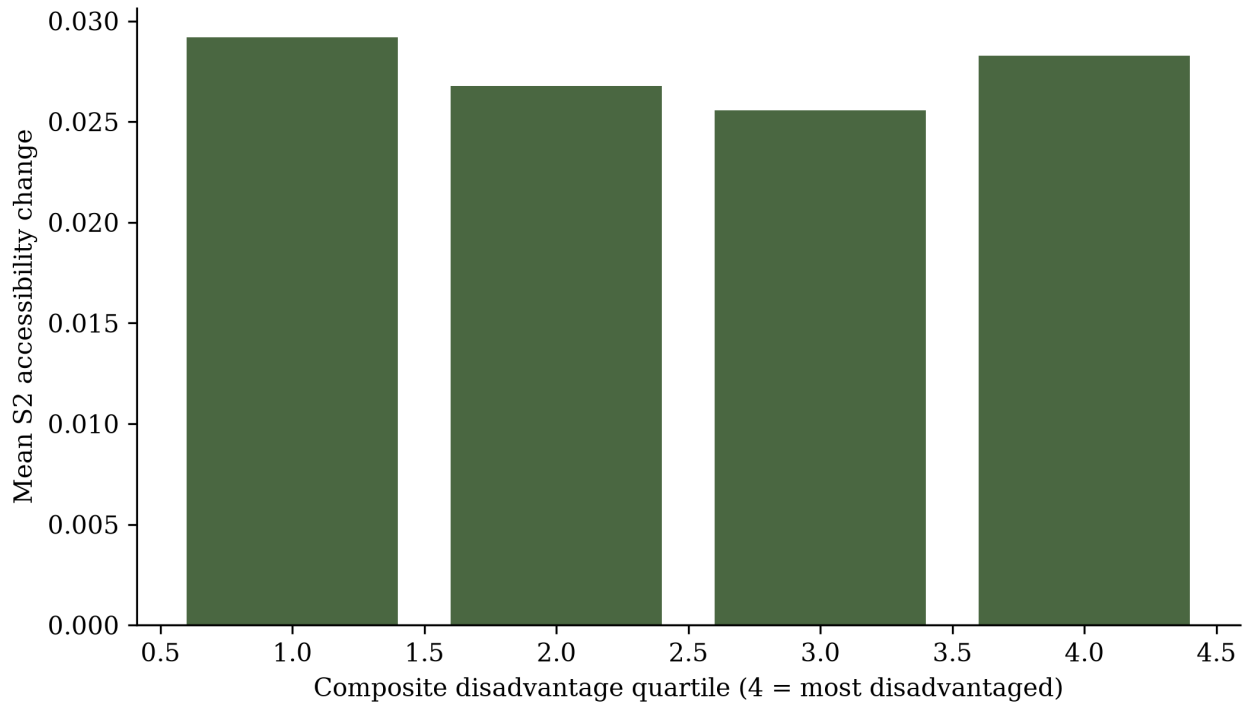
Prospective scenarios

Under **S2 (full announced 28×28)**, mean accessibility change is **0.0292** in Q1 vs **0.0283** in Q4 (Q4-Q1 = **-0.0009**). Spearman $\rho(\text{disadvantage, S2 } \Delta) = \mathbf{-0.113}$ ($p = 0.000$). OLS of S2 Δ on the composite index: $R^2 = \mathbf{0.0000}$ ($p = 0.889$).

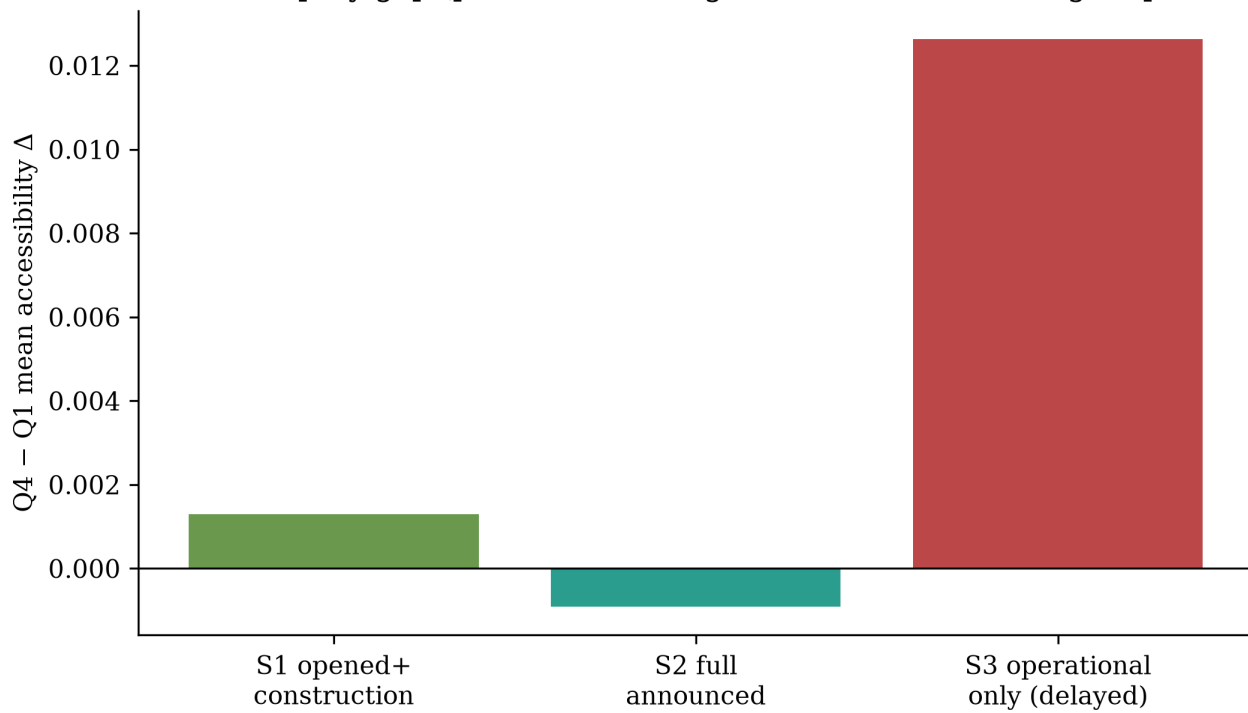
Prospective accessibility change if announced 28×28 points are delivered



Prospective access gains by disadvantage quartile (full announced scenario)

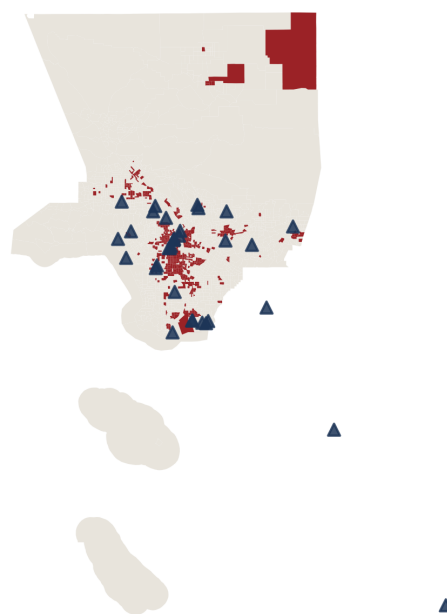


Scenario equity gap (positive = more gain in most-disadvantaged quartile)



Persistent deficits

Access deficit map: most-disadvantaged tracts (Q4)
outside 30-minute walk of any LA28 venue



Interpretation (placeholder): Venue geography and existing rail already place many higher-disadvantage tracts nearer to Games destinations than affluent periphery tracts. Announced 28×28 **point-proxied** gains do not show a strong progressive tilt toward the most disadvantaged quartile—an honest null/weak prospective result that motivates earlier, multimodal scenario design in Brisbane.

4. Policy interpretation

1. **Underserved:** Q4 tracts outside 30-minute walk access (deficit map) warrant community investment and first/last-mile attention, not only venue-adjacent spectacle.
 2. **Equity leverage:** Prioritize 28×28 elements that raise access for Q4 in sensitivity runs (BRT/ hubs over highway capacity where the metric is neighborhood access).
 3. **Do not equate proximity with benefit:** Highway projects and security footprints may impose burdens; see benefits/burdens memo.
 4. **Use before 2032:** Brisbane planners can run analogous scenarios **before** locking capital programs.
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5. Limitations

Planned infrastructure may change; accessibility ≠ utilization; announcements incomplete; point proxies for corridors; walk detour factors pending OSM; displacement not modeled; community listening protocol not yet executed. Full list: [METHODOLOGY .md](#).

6. Implications for Brisbane 2032 and Hickman

LA28 Equity Watch **identifies the pattern and proves the workflow**. It also exposes the gap: near-Games descriptive analysis is too late, and rail+detour methods are only a bridge to full multimodal routers and co-designed scenarios. Brisbane's planning-stage window + Hickman expertise supply the second phase — see [docs/RESEARCH_AGENDA_LA28_TO_BRISBANE.md](#).

Progression: LA28 identified the problem → spatial analysis demonstrated the pattern → limitations exposed the need for prospective multimodal modeling → Brisbane provides the planning-stage environment → Hickman provides transport-accessibility depth → improved framework returns comparatively to LA28.

Attribution

CalEnviroScreen 4.0 (OEHHA); HPI (Public Health Alliance / PHI via LA County Open Data); ACS & TIGER (U.S. Census Bureau); GTFS (LA Metro). Venues and 28×28 curated from public sources. Analysis: LA28 Equity Watch repository.